

I am Abhinav Arabelly, an AI engineer and data scientist currently based in San Jose, California. Over the past several years, my work has revolved around building intelligent systems that operate at scale, optimizing data pipelines, and designing machine learning solutions that are both technically rigorous and business impactful.

At Flywl, a cloud marketplace startup in California, I work as an AI Engineer where I focus on architecting production-grade AI systems. One of my primary contributions has been engineering an agentic Retrieval-Augmented Generation pipeline that indexed over 35,000 marketplace documents using vector storage and OpenAI-based indexing. This system significantly reduced internal decision-making time by approximately 90 percent by enabling intelligent search and contextual reasoning over unstructured product documentation. I also designed and implemented an automated data normalization and preprocessing framework that deduplicated more than 6,000 products across AWS, GCP, and Azure marketplaces. Ensuring cross-platform consistency required building a unique ID assignment architecture that scaled to 77,000 plus products while maintaining referential integrity. Beyond backend systems, I built a full-stack product catalog application using React and Flask, integrating fuzzy search mechanisms that improved product matching accuracy by 20 percent. The platform was deployed on AWS using EC2, S3, and vector storage systems, with PostgreSQL supporting activity logging for over 100 internal users. My role demands a blend of ML systems thinking, cloud architecture knowledge, and product-centric execution.

Prior to that, at the Wisconsin Institute for Discovery, I worked as an AI Research Engineer. My research focused on label-efficient supervised fine-tuning of large language models. I developed novel data selection strategies based on task-wise uncertainty estimation and clustering to identify the most informative training samples. Leveraging high-performance CUDA GPU clusters, I fine-tuned LLaMA-2 models on more than 100,000 samples across 1,800 plus NLP tasks. Our approach achieved an 80 percent reduction in training data requirements while maintaining or improving benchmark performance. I validated these improvements using AlpacaEval and statistical hypothesis testing, demonstrating consistent accuracy gains across diverse datasets. This research culminated in a main-track EMNLP 2025 publication on improving task diversity in label-efficient LLM fine-tuning. My work combined theoretical reasoning, experimental rigor, and large-scale engineering.

Earlier, at Asurion and Synechron Technologies in India, I worked as an AI and ML Engineer focusing on applied machine learning in industry settings. At Synechron, I contributed to fintech systems where reliability and precision were critical. I learned agile methodologies in real client-facing environments and worked on performance optimization for financial transaction systems. At Asurion, I fine-tuned large language models on over 30,000 customer support transcripts to build a generative AI-powered troubleshooting assistant. I streamlined transcript preprocessing pipelines using vectorized operations and regex transformations, reducing formatting overhead by 33 percent. I also conducted SQL-driven analysis over 500,000 plus mobile insurance records to identify root causes behind declining conversion rates, delivering actionable insights to sales and marketing teams. Through simulation-based agent allocation strategies, we forecasted an estimated 1.5 million dollar improvement in sales performance. I

built Tableau dashboards to democratize analytics across stakeholders, increasing data-driven decision adoption by 25 percent.

During my research tenure at Nanyang Technological University in Singapore, I developed an end-to-end neural speaker diarization system using PyTorch and SpeechBrain. I integrated CNN-based feature encoders and fine-tuned a WavLM transformer model on over 20 hours of real-world speech data, achieving a 5 percent reduction in diarization error rate. This experience strengthened my foundation in deep learning, speech systems, and research methodology.

Across all roles, my technical toolkit spans Python, SQL, R, C++, PyTorch, TensorFlow, Spark, Hadoop, AWS, GCP, LangChain, LangGraph, RAG architectures, vector databases, and MLOps tooling. My work style is analytical, iterative, and hypothesis-driven. I enjoy decomposing ambiguous problems into structured components and balancing experimentation with measurable impact.